



Double Glass N-Type Solar Panel

USER MANUAL V1.2



Model : CN450W

IEC 61215, IEC 61730

ISO 9001: 2023 Quality management systems

ISO 14001: 2023 Environmental management systems

ISO 45001: 2023 Occupational health and safety management systems

Thank you for purchasing this solar panel. Before using this product, please carefully read this manual. Once you begin using this product, you are considered to have read, understood, and agreed to the contents of this manual. Please strictly follow the instructions in this manual when using this product and pay close attention to the precautions.

We are not responsible for any direct or indirect damage or personal injury resulting from improper use of the solar panel, including but not limited to short circuits, overheating, insertion of foreign objects, unauthorized disassembly, physical impact, excessive mechanical pressure, or the use of incompatible or low-quality charge controllers.

****Please confirm in advance whether the parameters of this solar panel match the specifications of your equipment. Failure to meet these specifications may result in damage to the products.***

⚠ Product Use Precautions

1. Never attempt to disassemble, insert foreign objects into, damage, or modify the unit, as this may lead to unforeseen risks.
2. Keep the solar panel away from combustible materials during operation.
3. Avoid dropping, placing heavy objects on, or subjecting this unit to strong impacts.
4. Do not bend the solar panel. Handle with care during use to prevent collisions with the solar panel. Avoid touching it with sharp objects and refrain from forcefully striking the product.
5. Do not step on, stand on, or jump on the surface of the solar panel. Heavy or sharp objects may scratch the solar panel, causing small cracks and affecting output power or posing a risk of injury.
6. Do not pull the power cable hard to avoid the power cable from breaking or falling off.

Product Highlights

 Strengthen Mechanical Load Professional reinforced mechanical loads	 Safety Assurance Under Extreme Heat PV modules have excellent fire resistance
 More Energy Yield Higher energy production makes sure we always achieve your needs	 Anti -PID Excellent Anti-PID performances are ensured by optimized production process and material control



SMBB Super Multi Main Grid Technology

SMBB technology creates tougher, more efficient solar modules.



Parallel Connection Design

Innovative dual-module design ensures shading on one half won't impact the power output of the other.



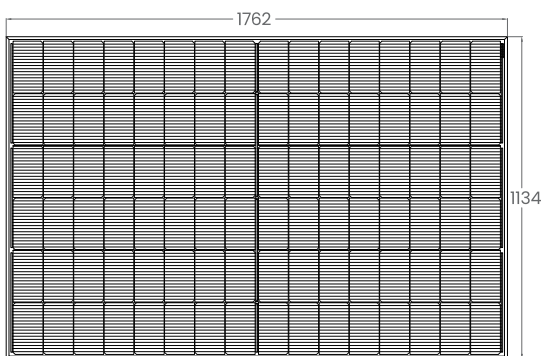
Double Glass Design

Unmatched reliability through cell encapsulation between two glass layers, ensuring superior durability and enhanced power output.

Product Dimensions

Model: CN 450W

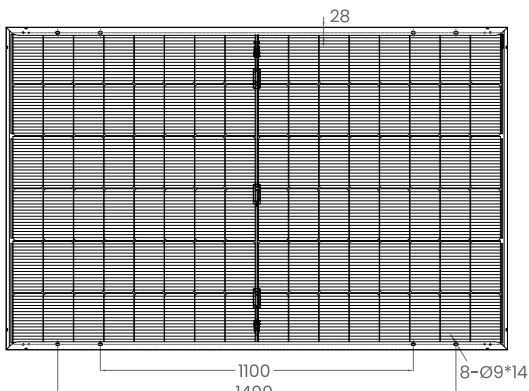
1762 × 1134 × 30 ±5mm



Front



Side



Back

Mechanical Characteristics(STC)

Model	CN450W
Maximum Power at STC (Pmax)	450W
Maximum Power Voltage at STC (Vmp)	30.08V
Maximum Power Current at STC (Imp)	14.97A
Open-circuit Voltage at STC (Voc)	35.03V
Short-circuit Current at STC (Isc)	16.05A
Cell Type	N-Type, 16BB, 182mm, Monocrystalline Silicon
Cell Efficiency	25.4%
Power Tolerance	0 ~ 3%
Maximum System Voltage	1500V DC
Maximum Series Fuse Rating	30A
First Year Degradation Rate	1%
Linear Degradation Rate	0.4%
Commitment to maintain 84.5% output	25 years

*STC(Standard Test Condition): Irradiance 1000W/m², Cell Temperature 25°C, Air Mass 1.5.

Specifications

Model	CN450W
Dimension	1762 × 1134 × 30 (±5mm)
	69.4 × 44.7 × 1.2 (±0.1in)
Weight	24.5 (±0.2kg)
	54 (±0.5lbs)
Glass	2.0+2.0mm Tempered Glass
Frame	Anodized Aluminum Alloy
IP Rating	IP68
Max. Wind Load/Snow Load	2400Pa / 5400Pa
Connection	Solar Connector

Temperature Coefficients

Model	CN450W
Temperature coefficients of Pmax	-0.29%/°C
Temperature coefficients of Voc	-0.25%/°C
Temperature coefficients of Isc	0.045%/°C
Operating Temperature (°C)	-40°C ~ 85°C

Product Maintenance

After long-term use, dust may adhere to the surface of the panel, blocking sunlight and reducing the cells ability to absorb sunlight. Regularly clean the solar panel by gently wiping it with a damp cloth. Do not use corrosive liquids to clean the solar panel.

Suitable Environments

To get the maximum power, the solar panel must be exposed to direct sunlight with no obstructions on its surface. Please use it under clear sunlight without clouds, trees, or buildings blocking the sunlight.

Warranty Policy

We provide a 10-year warranty service for materials and workmanship on this product. Within 10 years from the date of purchase, if the product experiences faults under normal usage, please apply for warranty with the purchase receipt.

The following situations are not covered by the warranty:

1. Cosmetic flaws caused during use.
2. Product malfunctions resulting from self-disassembly, repairs, or human-induced causes.
3. Damages caused by uncontrollable factors such as natural disasters, thunderstorms, accidents, and the like.

Warranty Application Process:

1. Please contact our customer service team, providing the purchase receipt and a detailed description of the issue.
2. Our customer service team will assist you in submitting the necessary information and documents.
3. After confirmation, we will provide you with an appropriate warranty solution.

If you have any questions or need assistance, please feel free to reach out to us via the following email: **callsun-service@outlook.com**

Specifications are subject to change without notice!

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